

BHABHA UNIVERSITY, BHOPAL

NH-12, Hoshangabad Road, Jatkhedhi, Bhopal-462026

BACHELOR OF COMPUTER APPLICATIONS (BCA)



DETAILED SYLLABUS (EFFECTIVE FROM JULY 2018)

**Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL**

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BACHELOR OF COMPUTER APPLICATIONS (BCA)

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DETAILED SYLLABUS

S.No.	Paper Code	Paper Name	Internal Evaluation	Theory	Practical	Total
1	BCA11	Fundamental Of Computer	25	50		75
2	BCA12	Communication Skill	25	50		75
3	BCA13	Programming Language C	25	50		75
4	BCA14	Discrete Mathematics	25	50		75
5	BCA15	Digital Computer Organization	25	50		75
6	BCA16	Accounting and Financial management	25	50		75
Practical						
7	BCA17	LAB-1(Programming language C)	20		30	50
8	BCA18	LAB-2(Office Management)	50		50	100
Total			220	300	80	600

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BCA11- Fundamental Of Computer

UNIT-I:

Introduction to computer: computer system characterization & capabilities, speed, accuracy, reliability, memory capability, repeatability. Computer hardware & software: block diagram of a computer, different types of software's. Types of computer: analog digital & hybrid, general and special purpose computers. Computer generations: characteristics of computer generations computer systems micros, minis & mainframes and pc introduction.

UNIT-II:

Computer organization introduction to input devices: categorizing input hardware. Storage devices : storage fundamentals, primary and secondary storage, data storage and retrieval methods-sequential, direct & indexed & sequential, tape storage and retrieval methods tape storage devices, characteristics & limitation, direct access storage for microcomputers- hard disks, disk cartridge, direct access storage devices for large computer systems, mass storage systems and optical disks cd rom.

UNIT-III

Data processing: data, data processing system, storing data, processing data. central processing unit: the microprocessor control unit, a l. u., register, buses main memory, main memory (RAM) for microcomputers, read-only memory .computer output: output fundamentals, hardcopy output devices, impact printers, non-impact printer's plotters, computer output microfilm/microfiche (com) system, softcopy output devices, cathodes ray tube and flat screen technologies.

UNIT-IV

Computer software system software system software versus application software, type of system. software's, introduction types of operating system programs, booting loader, diagnostic tests, operating system executive, bios, utility programs, file maintenance, language processors, assembler, compiler and interpreter, applications software: microcomputer software, interacting with system, trends in pc software, types of application software, difference between program and packages.

UNIT – V

Computer languages : computer programming languages, types of programming languages, generations of programming languages development low level versus high level language, machine ,advantages of using machine code, disadvantages of using machine code, assembly language, assembler, advantages of assemble languages, limitations of assembly languages. the need for assembly languages. High level languages.

TEXT & REFERENCE BOOKS:

1. COMPUTERS TODAY, BY S.K BASANDRA, GALGOTIA PUBLICATIONS.
2. FUNDAMENTALS OF INFORMATION TECHNOLOGY ALEXIS LEON & MATHEWS LEON.

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BCA12 - Communication Skill

UNIT-I

Language and Communication: Definition of Communication; Function and purpose of Communication; Process of Communication; Barriers of Effective Communication; Types of communication, Verbal communication, on-verbal communication; The Impact of Communication on Performance.

UNIT-II

Remedial English: Parts of Speech; Sentences; Subject-Verb Agreement; Active Voice and Passive Voice; Degrees of Comparison; Direct and Indirect Speech; Question Tags

UNIT-III

Oral Communications: Advantages and disadvantages of oral communication; Improving oral communication; One-to-One oral communication; Oral Presentations

UNIT-IV

Listening Skills: What is listening; Types of Listening; Barriers of Effective Listening; Strategies for Effective Listening; Semantic Markers; Listening to Complaints.

UNIT-V

Writing Skills: Note Taking; Paraphrasing; Elements of writing; Business Letter Writing; Other Business Communications, Types of Communication; Meetings; Memo; Circulars and Notices; Report Writing

TEXT & REFERENCE BOOKS:

- 1• A PRACTICAL ENGLISH GRAMMAR BY THOMSON AND MARTINET
- 2• ENGLISH GRAMMAR BY W.S.ALLEN
- 3• INTERMEDIATE ENGLISH GRAMMAR BY RA YMOND WILLIAMS
- 4• VOCABULARY BY MICHAEL MC CARTHU AND FELICITY O'DELL
- 5• ENGLISH GRAMMER BY JAYANTHI DAKSHINA MURTH

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BCA13 - Programming Language C

UNIT-I:

Introduction, Data Types and operators identifiers and keywords, constants, types of operators, type conversion, writing a C-Program, variable declaration, C-Statements, Input and Output functions

UNIT-II:

Control statement, conditional expressions if statement, if-else statement, case and switch statement, loop-statements: For loop, while loop, do while loop, Break, continue and go to statements

UNIT-III:

Functions and program structure, Function definition, Type of functions, local and global variables, scope of variable, multifunction programs, Recursive functions.

UNIT-IV:

Arrays Notation and declaration, initialization, multidimensional and character arrays, pointers, Declarations, Pointer arithmetic, pointers and functions.

UNIT-V:

Preprocessors and macros, Header files (brief introductions only), structures, Declarations, initialization and use of structures in a C-Program function and structures, Array of structures Arrays within a structures. Unions

TEXT & REFERENCE BOOKS:

- 1• HERBERT SHIELD, "COMPLETE REFERENCE C"
- 2• Y KANETKAR, "POINTERS THROUGH C "
- 3• Y KANETKAR, "TSR THROUGH C".
- 4• R.S SALARIA, "APPLICA TION PROGRAMMING IN C"

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BCA14 – Discrete Mathematics

UNIT-I:

Boolean algebra, Principle of Duality, Properties of Boolean Algebra, Inclusion Relation in Boolean Algebra, Boolean Sub algebra, Partial, Order Relations, Lower and Upper Bound Total order, Algebra of Propositions, Algebra of electric circuits: Switching Circuits Design of Simple Automatic Control System, Boolean Function of Fundamental Forms: Minimal Boolean Functions, Disjunctive Normal Form, Bool's Expansion Theorem, Logic Circuits Many Terminal Network, Some Definitions Related to Graph, Tree.

UNIT-II:

SET & OPERATIONS ON SETS: Union, Intersection, Disjoint Sets, Difference, Symmetric Difference, Complement Laws of Operations on Sets, Venn diagram, Generalized De Morgan's Laws, and Generalized Form of Distributive Laws. Cartesian product of sets and relations: Cartesian product of two sets Relation, Binary relation. Equivalence classes of equivalence sets. Properties of equivalence classes, Partition of a set. Function or mapping kinds of mapping. Some special types of mapping, Inverse function or Inverse mapping, Binary operations, Types of binary operations.

UNIT-III:

ELEMENTARY DIFFERENTIATION: Continuity and object of differential calculus. Related quantities, Variables, Function, functions of different kinds, Limits some important, expansions some theorems of limits. Some important limits, Right Hand and Left Hand limits. Continuity, Kinds of discontinuity, Properties of continuous functions. Basic concept of derivative of a function.

UNIT-IV:

ELEMENTARY INTEGRATION: Anti-derivative, indefinite integral, definite integral, Fundamental rules of integration, Standard formulae, Integration by substitution, Extended forms of fundamental formulae, some important integrals, integration by parts.

UNIT-V :

Partial differentiation, partial differentiation of higher order. Homogeneous functions, Total Differentiation, Differentiation of composite & Implicit functions. Changes of variables Taylors Theorem for several Variables.

TEXT & REFERENCE BOOKS:

- 1 J.P.Trembley & R.P.Manohar "Discrete Mathematical Structure with applications to Computer Science".
2. Kenneth H. Rosen-203 "Discrete Math & its Applications" 5th ed.
3. K.A. Ross and C.R.B. Writht "Discrete Mathematics ".
4. Bernard Kolman & Robert C. Busby "Discrete Mathematical Structures for Computer Science"

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BCA15 - Digital Computer Organization

UNIT-I

Data representation Data Types and Number Systems, Binary Number System, Octal & Hexa-Decimal Number System, Fixed Point Representation, 1's & 2's Complement, Binary, Arithmetic Operation on Binary Numbers, Overflow & Underflow, Floating Point Representation, Codes, ASCII, EBCDIC Codes, Gray Code, Excess-3 & BCD, Error Detection & Correcting Codes Binary Storage and Registers.

UNIT-II

Boolean algebra and digital logic circuits -Logic Gates, AND, OR, NOT,, NOR, NAND & XOR Gates and their Truth Tables, Boolean Algebra, Basic Definition and Properties, Basic Boolean Law's, Demorgan's Theorem, Minimization Techniques, K Map – Two, Three and More Variables maps, Sum of Product & Product of Sums, Don't care conditions.

UNIT-III

Combination Circuits - Half adder & Full adder, Full Subtractor, Full Subtractor and decimal adder, Code Conversion, Multilevel NAND and NOR Circuits, Decimal adder, decoders, Multiplexers and Demultiplexers.

UNIT-IV

Sequential logic- Flip-Flops - RS, D, JK & T Flip-Flop, Triggering in flip flops, Analysis of Clocked Sequential Circuits, State Reduction and Assignment, flip flop excitation tables, Design procedure and design of counters. Design with equations.

UNIT-V

Registers, Counters and the memory unit, Shift registers, Ripple counters and Synchronous counters, Inter-register Transfer, Arithmetic Logic and Shift Micro Operation, Conditional Control Statement, Instruction Codes, Processor organization, design of a simple computer.

TEXT & REFERENCE BOOKS:

1. DIGITAL LOGIC AND COMPUTER DESIGN BY MORRIS MANO
2. COMPUTER SYSTEM ARCHITECTURE BY MORRIS MANO

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BCA-16 Accounting and Financial management

UNIT – I

Introduction and Purpose of Accounting and Uses of Accounting Information, Basic Accounting Concepts, Accounting Structure, Process of Accounting, Journal, Ledger & Trial Balance based on Double Entry Book – Keeping.

UNIT – II

Valuation of Assets and Depreciation Methods: Straight Line Method, Diminishing Balance Method, Sinking Fund Method, Insurance Method and Aunty Method.

UNIT – III

Practical System of Accounting: Cash Book, Sales & Purchase of Goods, Bills of Exchange, Bank Reconciliation Statement.

UNIT – IV

Preparation of Financial Statements: Income Statement (Profit & Loss Account), Statement of financial Position (Balance Sheet) and Adjustment.

UNIT – V

Analysis of Financial Statements – Financial Ratio.

Reference Books:

1. Management Accounting By Hingorani, Grewal.
2. Financial Management by Khan & Jain.
3. Management Accounting by Shashi K Gupta, S Chand Publication.
4. Financial Accounts by Dr S M Shukla, Sahitya Bhavan Publications